

Multi-Expert Fusion with Confidence-Aware Abstention for Macro-Behavior Classification of Cow Vocalizations

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Automated interpretation of dairy cow vocalizations can support precision livestock farming by providing non-invasive indicators of welfare-relevant behavioural context; however, real-barn audio is highly variable, noisy, and uncertainty-prone. This paper presents a confidence-aware multi-expert late-fusion framework for macro-behaviour classification of cow vocalizations into nine operational groups (G0–G8). The framework combines three complementary specialists: a Call Signature Expert that models clip-level acoustic signatures using global descriptors, a Window Pattern Expert that captures short-time motifs from 3 s windows using frozen PANNs CNN14 embeddings, and an Event Histogram Expert that represents each clip as a Bag-of-Audio-Words distribution over learned acoustic events. Expert posterior probabilities are concatenated and fused using an XGBoost meta-classifier to produce the final macro-group distribution and confidence score. On 569 real-barn clips using a 455/114 train/test split, late fusion achieved 0.816 accuracy and 0.639 Macro-F1, outperforming all individual experts. Additional analyses show that the 3 s non-overlapping configuration provided the strongest fusion Macro-F1 among tested window settings, while confidence-thresholded abstention improved reliability by enabling a controllable coverage–accuracy trade-off. Calibration analysis further showed that the fusion confidence was useful but imperfect (ECE = 0.100; multiclass Brier = 0.294), supporting its use as a practical deployment-oriented confidence signal rather than a full uncertainty model. These results indicate that integrating global, local, and distributional acoustic evidence with selective prediction is an effective and pragmatic strategy for macro-behaviour recognition in noisy farm environments.

Keywords - precision livestock farming, bovine bioacoustics, cow vocalizations, macro-behaviour classification, ensemble learning, late fusion, confidence estimation, abstention, Bag-of-Audio-Words, acoustic embeddings.

I. INTRODUCTION

Precision livestock farming increasingly depends on continuous sensing to support early, non-invasive welfare monitoring [1]–[3]. Bioacoustic monitoring is particularly attractive because cattle express physiological and emotional states through vocalizations, often before visible behavioral symptoms emerge, positioning vocal cues as practical biomarkers for welfare assessment in commercial settings [1], [4], [5].

Despite rapid progress in machine learning for animal sound analysis, deployment in dairy barns remains challenging. Commercial barns are acoustically complex environments containing background machinery, overlapping calls, and routine human activity; as a result, models developed on clean or controlled recordings often degrade

when applied to field audio embedded in heterogeneous soundscapes [6]–[8]. In addition, many systems still produce forced labels even when the acoustic evidence is ambiguous, which can reduce user trust and practical utility in decision support settings. Selective prediction, or abstention, addresses this problem by allowing the model to defer uncertain cases rather than returning potentially unreliable outputs [9]–[13].

Recent work has established important prerequisites for deployment-oriented research. A FAIR-compliant bovine vocalization resource has been reported with 569 curated clips collected from three commercial dairy farms under ecologically realistic barn conditions [14]. In parallel, MooLogue has demonstrated a practical pathway for farmer- and student presenting predicted labels and associated confidence values as user-facing metadata within a curated application workflow [15].

This paper addresses the remaining gap: stable macro-behavior inference from real barn audio with explicit confidence handling. The target task is macro-level behavioral grouping into nine operational classes: G0 Social/Group Contact, G1 General Discomfort, G2 Handling Distress/Pain Context, G3 Feeding/Rumination, G4 Feed & Water Frustration, G5 Estrus/Mating, G6 Water Quality Response, G7 Feed Anticipation/Empty Feeder, and G8 Social Distress/Herd Context. In this setting, a reliable system must not only predict a macro group but also indicate when evidence is insufficient and abstain from prediction [9]–[13].

To address this problem, we present a confidence-aware multi-expert late-fusion framework that integrates complementary acoustic evidence through three specialized experts: a Call Signature Expert that captures clip-level acoustic signatures, a Window Pattern Expert that models short-time patterns using frozen PANNs CNN14 embeddings [16], and an Event Histogram Expert that summarizes each clip as a distribution of learned acoustic events in a Bag-of-Audio-Words representation [17]. Their posterior probabilities are fused into a final macro-behaviour prediction using an XGBoost meta-classifier, and a confidence-based abstention mechanism enables selective prediction by trading coverage for reliability [9]–[13]. We further analyze the proposed design through window-size sensitivity, per-class error patterns, calibration, and threshold-based abstention behaviour, providing a more explicit assessment of when the system is reliable and when it should defer prediction.

The main contributions of this paper are:

- A three-expert fusion architecture (Call Signature Expert, Window Pattern Expert, Event Histogram Expert) designed for macro-behavior classification (G0–G8) in noisy barn audio.
- A confidence-aware abstention strategy that supports selective prediction, improving trustworthiness by allowing the model to defer uncertain cases [9]–[13].
- An empirical study on 569 ecologically realistic dairy-barn vocalization clips showing that fusion improves balanced macro-level recognition over all individual experts, with additional analyses of window-size choice, class imbalance effects, and per-class error patterns.

II. RELEVANT WORK

Machine learning for livestock vocalization analysis has progressed from handcrafted acoustic descriptors toward deep representations and multimodal sensing pipelines, with increasing emphasis on translating vocal patterns into welfare-relevant behavioural context [1], [6], [8], [14]. However, many studies still rely on relatively clean or controlled recordings, whereas commercial dairy barns contain persistent background machinery, overlapping calls, and human activity that degrade performance under real deployment conditions [6]–[8]. Recent work has therefore emphasized ecologically realistic datasets and reproducible pipelines to support more robust field-oriented modelling [7], [14].

On the deployment side, user-facing systems increasingly require interpretable outputs and confidence communication. MooLogue operationalizes this need by storing predicted labels with confidence values during curation and presenting them as metadata within an educational app interface, reinforcing that confidence is not only a modeling concern but also a communication requirement for end users [15]. In parallel, selective prediction (reject-option) methods provide a principled mechanism for abstaining on uncertain inputs, reducing error at the cost of coverage; modern formulations extend these ideas to deep models through confidence thresholding and bounded-risk selective classification [9]–[13]. The approach presented in this paper builds on these directions by combining multi-resolution expert evidence with confidence-aware abstention for macro-behavior inference in real barn audio.

III. METHODOLOGY

A. Data preparation and macro labels

The study targets macro-behaviour classification of cow vocalizations recorded under real commercial barn conditions [14]. To improve robustness and reduce ambiguity from sparse fine-grained labels, the available vocalization clips are consolidated into nine macro-behaviour groups (G0–G8), each representing an operationally meaningful behavioural context.

All recordings are standardized to WAV (mono, 32 kHz) to ensure consistent decoding and feature extraction. Clip segmentation follows a context-aware workflow in which short calls are extracted with padding when needed (2–3 s of padding around candidate events) to preserve surrounding context. To ensure biologically meaningful windows, active-centered selection is applied when needed: vocal activity is detected (e.g., using `librosa.effects.split()`), and a ~3s segment

is centered on the active region with 30–40% context retained [18]. The evaluation uses 569 curated clips, split into 455 training clips and 114 test clips. For the window-based experts, each clip is segmented into non-overlapping 3 s windows in the primary configuration, producing 3276 training windows and 848 test windows. Additional window settings are evaluated separately to examine sensitivity to window length and overlap, but the configuration uses the 3 s non-overlapping design because it yielded the strongest fusion Macro-F1. This macro-level formulation is intended to improve behavioural stability under real-barn acoustic variability while preserving categories that remain interpretable for downstream welfare analysis.

B. Call Signature Expert (clip-level global acoustics)

The Call Signature Expert is designed to capture the global acoustic identity of a vocalization (i.e., the overall timbre/energy/noisiness profile of a clip), which is often stable even when short segments are corrupted by barn noise. This expert operates at the full-clip level and provides a compact, interpretable representation of each call through a set of 34 clip-level acoustic descriptors, including MFCC statistics [19], spectral centroid/roll-off/bandwidth, RMS/energy measures, zero-crossing rate, and duration-related descriptors (computed using standard audio analysis routines [18]). These descriptors have clear physical meaning; for example, spectral centroid and roll-off relate to “brightness/noisiness,” while MFCCs capture vocal timbre patterns relevant for call-type differentiation [19].

Concretely, features are normalized (e.g., `StandardScaler`) and a balanced Logistic Regression classifier is trained to mitigate macro-group imbalance using standard ML tooling [20]. The model outputs a 9-way probability vector $p_A \in R^9$ over G0–G8 for each clip, functioning as a stable “global prior” for the late-fusion stage. Because it summarizes the entire clip, this expert is robust to local perturbations but can miss short-term micro-events (brief pulses, bursts, or transient harmonics) that distinguish acoustically similar macro contexts.

C. Window Pattern Expert (local temporal patterns)

The Window Pattern Expert is designed to capture short-time acoustic motifs that may be diluted when an entire clip is summarized into a single global descriptor. In the configuration, each clip is partitioned into non-overlapping 3 s windows. For each window, a frozen PANNs CNN14 embedding is extracted as a transferable representation learned from large-scale audio data [16]. The resulting embedding dimension is 2048, and the encoder is used only as a fixed feature extractor without task-specific fine-tuning.

A lightweight class-balanced Logistic Regression classifier [20] is trained on the window embeddings to predict macro-level posterior probabilities for individual windows. Clip-level expert output $p_B \in R^9$ is then obtained by averaging the window-level posterior vectors across all windows belonging to the clip.

This expert is intentionally “local”: it is expected to be noisier per window than clip-level features, but it contributes information that the Call Signature Expert cannot capture—namely, short-duration patterns and local transitions. The fusion model is therefore able to treat p_B as evidence for macros that are characterized by distinct short-time signatures, even when full-clip cues overlap across contexts. Sensitivity

analyses for window length and overlap are presented in Section IV to justify the 3 s configuration used.

D. Event Histogram Expert (distribution of acoustic events)

The Event Histogram Expert models a clip as a distribution of recurring acoustic events rather than as a single summary or a strict temporal sequence. This is motivated by the observation that some macro behaviors are expressed through repeated motifs (e.g., recurring contact-like calls, repeated frustrated bursts) and are better captured through frequency-of-occurrence statistics than by a single global descriptor.

This expert uses the same 3-second window embeddings as the Window Pattern Expert, pools all training-window embeddings, and learns an unsupervised codebook via KMeans clustering [21]. In the reported configuration, the codebook contains 64 audio “words.” Each window is assigned to its nearest codeword, and each clip is represented by a 64-dimensional histogram (Bag-of-Audio-Words) indicating how often each acoustic event type occurs within the clip [17]. The histogram becomes a stable clip-level vector that compresses variable-length window sequences into a fixed-length representation that is less sensitive to local misalignment or missing windows [17].

A discriminative classifier (XGBoost) is trained on these histograms to output the expert probability vector $p_C \in R^9$ over macro groups [22]. This expert typically complements the other two by contributing distributional evidence: even if individual windows are ambiguous, the overall composition of event types across a clip can be strongly informative for certain macros (e.g., consistent “feeding-like” motifs versus mixed social-distress patterns).

Together, the three experts provide complementary evidence—global call identity (Call Signature), localized short-term motifs (Window Pattern), and event-composition statistics (Event Histogram), which is explicitly the rationale for adopting late fusion rather than a single monolithic representation.

E. Late Fusion of expert decisions

Late fusion is applied so that each expert contributes an interpretable probability estimate, and a meta-learner learns when to trust each view. For a given clip, the three experts produce:

- $p_A \in R^9$: Call Signature Expert probabilities
- $p_B \in R^9$: Window Pattern Expert aggregated probabilities
- $p_C \in R^9$: Event Histogram Expert probabilities

These are concatenated into a 27-dimensional fusion feature vector:

$$\mathbf{z} = [p_A \mid p_B \mid p_C] \in R^{27}$$

A multiclass XGBoost fusion model maps $\mathbf{z}$ to a final 9-way probability vector p_F , learning which expert is most informative for each macro group [22]. This design allows the fusion stage to exploit complementarity across experts: some macro groups are better captured by stable clip-level signatures, while others benefit more from local temporal patterns or recurring event composition.

F. Confidence-aware abstention (selective prediction)

In field deployment, forced predictions on ambiguous barn audio can reduce trust and create false alarms. To address this, the fusion output is paired with a confidence score:

$$c = \max_k p_F[k].$$

Given an abstention threshold τ , the decision rule is:

- if ($c < \tau$), abstain (no label returned)
- else output $\arg \max_k p_F[k]$.

This mechanism enables selective prediction by allowing the system to defer uncertain inputs rather than forcing a label [9]–[13]. The same confidence definition (max probability) is used for the fusion model to generate an abstention curve. Here, abstention is analyzed by sweeping τ and measuring coverage, accuracy, Macro-F1, and risk, while calibration quality is summarized separately using expected calibration error and multiclass Brier score. Accordingly, the confidence score is treated as a practical deployment-oriented reliability signal rather than as a complete uncertainty model.

G. Algorithm: Confidence-aware late fusion inference

1. Extract clip-level acoustic features $\rightarrow$ compute p_A using Call Signature Expert [18], [19], [20].
2. Segment the clip into 3 s windows, extract frozen CNN14 embeddings for each window, infer window-level posteriors, and average them to obtain p_B [16], [20].
3. Assign the same window embeddings to a 64-word KMeans codebook, build a clip-level Bag-of-Audio-Words histogram, and infer p_C using Event Histogram Expert [17], [20], [21], [22].
4. Concatenate $[p_A \mid p_B \mid p_C] \rightarrow$ infer the final posterior p_F using multiclass XGBoost late fusion [22]
5. Compute $c = \max(p_F)$. If ($c < \tau$), abstain; otherwise output $\arg \max(p_F)$ [9]–[13]

IV. EXPERIMENTS AND RESULTS

A. Dataset and Experimental Setup

Experiments were conducted on 569 curated real-barn cow vocalization clips [14], consolidated into nine macro-behaviour classes (G0–G8). The evaluation uses a 455/114 train/test split at the clip level. For the two window-based experts, the configuration segments each clip into non-overlapping 3 s windows, yielding 3276 training windows and 848 test windows.

The evaluated system consists of three complementary experts: (i) a Call Signature Expert operating on full-clip acoustic descriptors, (ii) a Window Pattern Expert operating on frozen CNN14 embeddings extracted from 3 s windows, and (iii) an Event Histogram Expert operating on Bag-of-Audio-Words histograms derived from the same window embeddings. Each expert produces a 9-way posterior vector, and the final prediction is obtained by concatenating these posteriors into a 27-dimensional fusion feature and training an

XGBoost meta-classifier. The fusion model is evaluated on a fully held-out test set with no overlap with expert training data.

The experiments further examine fusion ablations, window-size and overlap sensitivity, calibration behaviour, threshold-based abstention performance, and class-imbalance effects.

B. Metrics

Performance is primarily reported using Accuracy and Macro-F1. Macro-F1 is emphasized because macro-behaviour recognition remains strongly class-imbalanced, and a model with high overall accuracy may still underperform on rare but welfare-relevant behavioural groups [14]. To evaluate selective prediction, abstention is assessed using coverage together with retained-set accuracy, Macro-F1, and risk under confidence thresholding. Confidence quality is further summarized using expected calibration error (ECE) and multiclass Brier score.

C. Quantitative Results

Table I summarizes the configuration results. Among the individual experts, the Event Histogram Expert achieved the strongest standalone performance (Accuracy 0.702, Macro-F1 0.550), indicating that the distribution of recurring acoustic events provides robust macro-level evidence in noisy barn audio. The Call Signature Expert provided a stable clip-level baseline (Accuracy 0.684, Macro-F1 0.433), while the Window Pattern Expert achieved Accuracy 0.614 and Macro-F1 0.467 after clip-level aggregation of window posteriors.

TABLE I. PERFORMANCE OF INDIVIDUAL EXPERTS AND LATE FUSION

Model	Representation	Accuracy	Macro-F1	Weighted-F1	Brier
Call Signature Expert (A)	Clip descriptors	0.684	0.433	0.688	0.502
Window Pattern Expert (B)	3 s window embeddings → clip aggregation	0.614	0.467	0.600	0.662
Event Histogram Expert (C)	Window-codeword histogram	0.702	0.550	0.696	0.422
Late Fusion (A+B+C)	Concatenated expert posteriors	0.816	0.639	0.800	0.294

The late-fusion model achieved the best overall result, reaching 0.816 accuracy and 0.639 Macro-F1 on the 114-clip test set. This improvement over all individual experts supports the central hypothesis of the paper: macro-behaviour recognition benefits from integrating complementary global, local, and distributional acoustic evidence rather than relying on a single representation.

D. Fusion Ablation and Window Sensitivity

Fusion ablation analysis shows that combining all three experts yields the strongest overall performance. As shown in Table II, the best two-expert combination was A+B (Accuracy 0.798, Macro-F1 0.628), while the full A+B+C

fusion remained strongest overall (Accuracy 0.816, Macro-F1 0.639). In contrast, A+C and B+C produced lower Macro-F1 values, indicating that the Window Pattern Expert contributes useful complementary local evidence even though its standalone performance is lower than that of the Event Histogram Expert.

TABLE II. FUSION ABLATION RESULTS ON THE 3 S CONFIGURATION

Fusion variant	Accuracy	Macro-F1	Weighted-F1	Brier
A+B	0.798	0.628	0.788	0.301
A+C	0.763	0.493	0.750	0.341
B+C	0.737	0.478	0.716	0.401
A+B+C	0.816	0.639	0.800	0.294

Sensitivity to window design is shown in Fig. 1. Among the tested settings, the 3 s non-overlapping configuration yielded the strongest fusion Macro-F1, while the 3 s configuration with 50% overlap achieved the highest fusion accuracy but lower Macro-F1. The 2 s and 4 s non-overlapping variants produced lower balanced fusion performance. These results support the use of 3 s non-overlapping windows as the primary configuration for macro-behaviour classification.

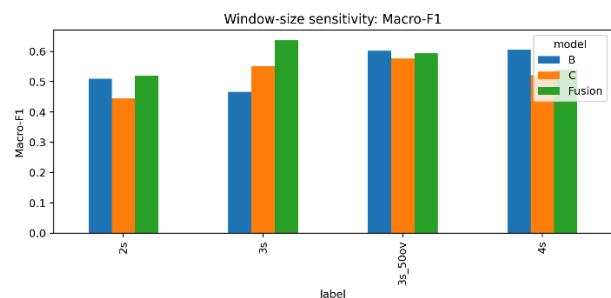


Figure 1: Window-size sensitivity analysis for B, C, and Fusion using Macro-F1. The 3 s non-overlapping configuration provided the strongest balanced fusion performance.

E. Confidence, Calibration, and Abstention

The usefulness of the fusion confidence score for selective prediction is summarized in Fig. 2. Without abstention, the fusion model achieved 0.816 accuracy and 0.639 Macro-F1 on the full 114-clip test set. As the abstention threshold increased, accuracy improved at the cost of reduced coverage, demonstrating a controllable coverage–reliability trade-off. For example, at $\tau = 0.5$ the model retained 96.5% coverage with 0.827 accuracy, 0.651 Macro-F1, and risk 0.173 on retained predictions; at $\tau = 0.7$ coverage decreased to 86.8% with 0.879 accuracy, 0.660 Macro-F1, and risk 0.121; and at $\tau = 0.9$ coverage further decreased to 67.5% with 0.935 accuracy, 0.605 Macro-F1, and risk 0.065.

Calibration quality was also evaluated for the fusion model. The model achieved a top-label expected calibration error of 0.100 and a multiclass Brier score of 0.294. These values indicate that the confidence score is useful for threshold-based abstention, although it should be interpreted as a practical reliability signal rather than a complete uncertainty estimator.

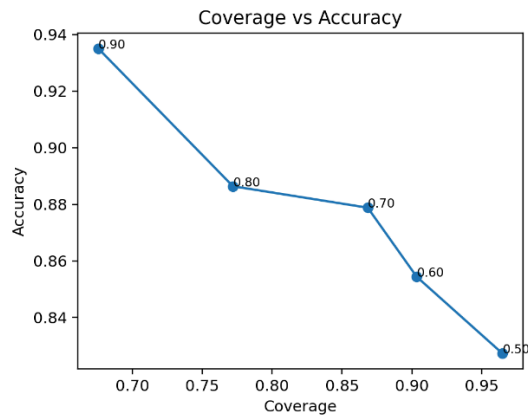


Figure 2: Coverage–accuracy trade-off of the late-fusion model under confidence-thresholded abstention. Higher thresholds reduce coverage while increasing accuracy on retained predictions.

F. Per-Class Performance and Class-Imbalance Effects

Per-class analysis of the fusion model helps explain why Macro-F1 remains substantially lower than the headline accuracy despite strong overall performance. The model performed best on well-supported and acoustically distinctive macro groups (shown in Table III), including G3 Feeding/Rumination ($F1 = 0.938$), G5 Estrus/Mating ($F1 = 0.926$), G7 Feed Anticipation/Empty Feeder ($F1 = 0.773$), and G2 Handling Distress/Pain Context ($F1 = 0.759$). In contrast, performance remained weak on low-support minority classes such as G0 Social/Group Contact ($F1 = 0.286$), G4 Feed/Water Frustration ($F1 = 0.400$), and especially G1 General Discomfort ($F1 = 0.000$; support = 2). The single test instance for G6 Water Quality Response was classified correctly, but that result should not be overinterpreted because of the extremely small support.

These classwise patterns indicate that the remaining errors are driven primarily by residual overlap among behaviourally adjacent macro groups and limited support for minority classes, rather than by uniform failure across all categories. Additional imbalance analysis showed that class balancing strategies affected experts differently: SMOTE improved the Window Pattern Expert substantially, but did not improve the Call Signature Expert or the Event Histogram Expert. This suggests that imbalance mitigation is representation-dependent in this problem setting and that no single strategy is uniformly optimal across all experts.

TABLE III. PER-CLASS PERFORMANCE OF THE LATE-FUSION MODEL

Macro class	Precision	Recall	F1	Support
G0	0.333	0.250	0.286	4
G1	0.000	0.000	0.000	2
G2	0.647	0.917	0.759	12
G3	0.882	1.000	0.938	30
G4	0.500	0.333	0.400	3
G5	0.893	0.962	0.926	26
G6	1.000	1.000	1.000	1

G7	0.850	0.708	0.773	24
G8	0.778	0.583	0.667	12

G. Computational Footprint and Deployment Considerations

The proposed framework is lightweight at the model level despite using three complementary experts. In the configuration, the learned model sizes were approximately 0.005 MB for the Call Signature Expert, 0.142 MB for the Window Pattern Expert classifier, 1.918 MB for the Event Histogram Expert, and 1.846 MB for the fusion model. The final fusion input remains compact (27 dimensions), while the Event Histogram branch compresses variable-length window structure into a fixed 64-dimensional representation. The full pipeline runtime for the configuration was 135.12 s for the reported training-and-evaluation workflow on the available dataset size. These characteristics suggest that the proposed system is more deployment-friendly than heavier end-to-end retraining approaches, although broader runtime benchmarking across hardware settings remains future work.

V. DISCUSSION

The results show that macro-behaviour recognition from real-barn cow vocalizations benefits from combining complementary acoustic views rather than relying on a single representation. Among the individual experts, the Event Histogram Expert provided the strongest standalone balanced performance, while the full late-fusion model achieved the best overall result across accuracy, Macro-F1, weighted-F1, and Brier score. This indicates that the proposed system does not improve performance through a single dominant branch alone; instead, the fusion stage benefits from integrating stable clip-level cues, short-time temporal patterns, and clip-level event composition. The ablation results further support this interpretation, with the full A+B+C configuration outperforming all two-expert variants.

A second key finding is that the temporal design of the window-based experts matters. The 3 s non-overlapping configuration provided the strongest fusion Macro-F1 among the tested settings, while a 3 s overlapping configuration slightly increased accuracy but reduced balanced performance. This suggests that, for the present dataset, 3 s windows offer a useful compromise between retaining localized acoustic structure and preserving enough context for macro-behaviour interpretation. The result also helps explain why the Window Pattern Expert remains valuable in fusion even though its standalone performance is lower than that of the Event Histogram Expert.

The confidence-aware prediction layer strengthens the practical value of the framework by allowing the system to abstain on uncertain clips rather than forcing a label in all cases. The observed coverage–accuracy trade-off shows that higher confidence thresholds improve reliability on retained predictions, which is important in barn environments where overlapping sounds, background machinery, and sparse minority classes can make some clips inherently ambiguous. At the same time, the calibration results indicate that maximum posterior confidence is useful but not perfectly calibrated, so it should be interpreted as a practical reliability signal rather than a complete uncertainty estimate. We adopt

maximum posterior probability as a simple and deployment-aligned confidence proxy; more advanced calibration strategies remain part of future work.

Per-class results also provide important context for interpreting the fusion performance. Strong results on classes such as Feeding/Rumination, Estrus/Mating, and Feed Anticipation/Empty Feeder indicate that the framework is effective when a macro group is both sufficiently supported and acoustically distinctive. In contrast, low-support classes such as General Discomfort and Feed/Water Frustration remain challenging, which explains why Macro-F1 remains lower than the headline accuracy. The imbalance analysis further shows that mitigation strategies are representation-dependent: SMOTE improved the Window Pattern Expert substantially, but did not improve the Call Signature Expert or the Event Histogram Expert.

Despite these strengths, several limitations remain. First, the study is based on 569 curated clips from three commercial farms, which limits generalization across broader acoustic environments, management conditions, and animal populations. Second, the current study does not include benchmarking on external public datasets or direct comparison with stronger modern baselines such as AST- or YAMNet-style audio classifiers. Third, the confidence score is based on maximum posterior probability and therefore should not be interpreted as a full uncertainty model. Finally, although the present analyses strengthen the empirical justification of the framework, broader validation with more diverse recordings, external benchmarks, richer augmentation strategies, and more advanced uncertainty estimation remains important future work.

VI. CONCLUSION

This paper presented a confidence-aware multi-expert late-fusion framework for macro-behaviour classification of cow vocalizations recorded in real barn environments. The approach integrates three complementary specialists - Call Signature Expert for global clip-level acoustics, Window Pattern Expert for local 3 s window patterns using frozen CNN14 embeddings, and Event Histogram Expert for Bag-of-Audio-Words event composition and combines their posterior probabilities through an XGBoost fusion model.

Experiments on 569 curated clips showed that the late-fusion model achieved the strongest overall performance, reaching 0.816 accuracy and 0.639 Macro-F1 on the 114-clip test set, outperforming all individual experts. Additional analyses showed that the full three-expert fusion outperformed all partial fusion variants, that 3 s non-overlapping windows provided the strongest balanced fusion performance, and that confidence-thresholded abstention enabled a controllable trade-off between prediction coverage and reliability. Overall, the results indicate that combining global, local, and distributional acoustic evidence is a practical strategy for robust macro-behaviour recognition in noisy barn audio. While broader validation remains necessary, the framework provides a compact and interpretable step toward more reliable bioacoustic decision support for precision livestock farming.

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